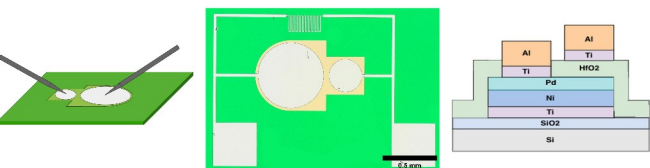


Introduction

MIM capacitors consist of 2 metal layers separated by a thin, insulating dielectric layer that stores energy in an electric field. Adding serpentine resistors to MIM capacitors in series or parallel creates RC circuits which are widely used in industry for Signal filtering and energy storage.

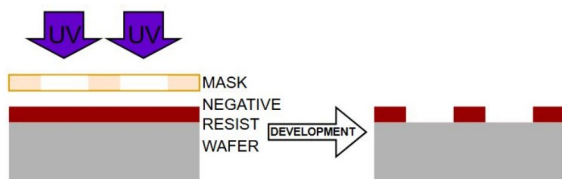


Statistical Process Control (SPC) is widely used in the industry to monitor manufacturing processes and ensure fabrication consistency. Our team focuses on lithography data, such as capacitor area and photoresist uniformity. The design for our team includes a 2.5 nF capacitor and 780 ohm resistor in a parallel configuration

Lithography

Process Overview

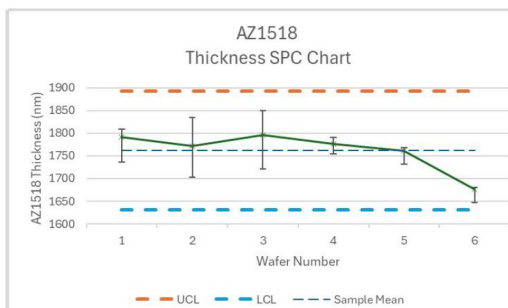
- Photolithography uses ultraviolet (UV) light to transfer patterns from a mask onto a photoresist layer coated on the wafer surface.
- UV exposure chemically alters the photoresist, the exposed regions remain after development.
- The development process removes unexposed regions, forming the desired patterned features on the wafer.
- Team evaluated the uniformity of lithography across wafers, assessed the quality, and consistency of the fabricated wafers. The expected Photoresist thickness was set at 1800 nm, and the expected capacitor area was 0.28 mm²



SPC Data Collection

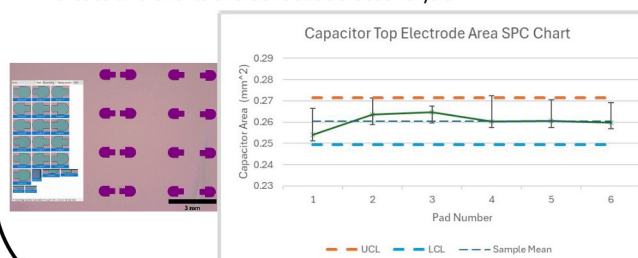
Film Thickness

- Utilized the Filmetrics F40-UV thin-film analyzer to measure thickness and optical constants.
- Analyzed spectral light reflections across multiple wavelengths to determine precise resist thickness.
- Verified that resist profiles remained within the defined process window, ensuring consistent and repeatable lithography results.



Area Measurement

- The Keyence VHX 7000 optical microscope was used to collect area measurements for the capacitor pads to ensure reliability of exposure step.
- A custom profile was created in the microscope software to auto-detect capacitor pads. These measurements were then used to create SPC Charts and conduct defect analysis

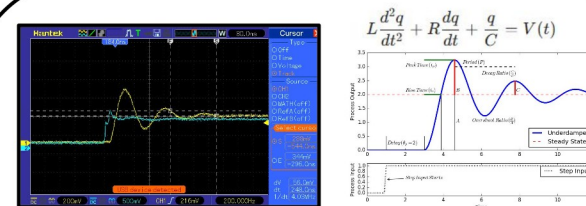


Acknowledgements

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1) APMonitor. (n.d.). Second order applications. <https://apmonitor.com/pdc/index.php/Main/SecondOrderApplications>

Characterization Results



Current Data Analysis

- Above on the left shows an oscilloscope capture of the input response of a 100Hz, 1Vp-p, square waveform. While other teams RC circuits were characterized by exhibiting classical first-order step responses governed by the formula $\tau = RC$, our parallel architecture demonstrates a distinctly underdamped, second-order response.
- As observed in the captured oscilloscope waveform, the high-frequency edge of the input pulse interacts with parasitic inductance inherent to the serpentine geometry and on-wafer probing environment.
- This resulting inductance transforms the ideal parallel RC network into an RLC circuit governed by the equation and figure shown in the top right. Consequently, rather than yielding a simple time constant extraction, the measurement reveals a pronounced ringing at the system's damped natural frequency: $\omega_d = \omega_n \sqrt{1 - \zeta^2}$
- This high-frequency oscillation highlights the effects of natural component impedance resulting in reactance that affects the circuit's response to AC waveforms.

Conclusion / Future Work

- Filter and process feature-area measurement data to directly compare theoretical and actual values.
- Probe RC circuits and MIM capacitors to measure capacitance, and measure resistance under a DC source.
- Model the measured impedance as equivalent Rs-Cs series or parallel circuits.
- Calculate the ideal theoretical impedance of the integrated structures and compare it against measured result